Title: Retail Sales Analytics Dashboard and Data Integration Project

An Internship Report

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY in Computer Science and Engineering

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Batch: 2026-2027

CANDIDATE’S DECLARATION

I, Manan Sharma do hereby declare that the internship report titled **“Retail Sales Analytics dashboard and data integration project ”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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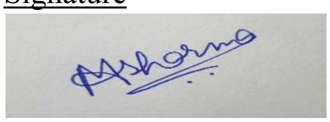

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name Manan Sharma

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Internship Completion Certificate

Eagletfly Solutions

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Manan Sharma** has been working with us as a **Data Analyst Intern 6th June'26 to 6th July'26** During this tenure he has completed his project successfully.

This is further certified that his progress is excellent and as per the industry standards.

We wish him all the best for his future assignments.

For **Eagletfly Solutions**

Authorized Signatory

Contents

Abstract

This internship project involved conducting a complete analysis of retail sales data using SQL, Python, and an online AI-based dashboard tool. The process began with uploading and merging the customers and transaction details datasets using SQL JOIN operations to create a unified data structure. The merged dataset was then imported into Python for cleaning, preprocessing, and exploratory data analysis, which included handling missing values, removing duplicates, correcting data types, and generating statistical summaries. Through EDA, several key insights were discovered, such as wallet payments generating the highest profit and cash payments receiving the highest discounts. Store_C performed as the most profitable store, while Store_D showed the highest discount distribution. Daily trends revealed fluctuations in customer activity and store performance. The insights were visually represented using AI-generated dashboards, allowing for clear and interactive interpretation. Overall, the internship strengthened practical skills in data integration, data analysis, and visualization in a real-world context.

1. INTRODUCTION

1.1 Overview

In modern data driven organizations, transactional datasets serve as the backbone for strategic insights and decision making. Businesses rely heavily on accurate, timely, and well processed data to understand customer behavior, store performance, profitability, and discount patterns. The internship at EagletFly Solutions provided a comprehensive opportunity to work with real world sales datasets and understand how data engineering and analysis can support business intelligence. The project included uploading CSV datasets into SQL, merging them using JOIN operations, cleaning and transforming the combined dataset using Python, and generating analytical dashboards using Power BI. Each phase of the workflow played a critical role in ensuring data quality, consistency, and usability for further analysis.

1.2 Objectives

The main objectives of the system are:

- To upload and store multiple datasets in SQL for efficient management.
- To perform SQL JOIN operations to merge customer and transaction datasets.
- To clean inconsistent, missing, and duplicate data using Python in Jupyter Notebook.
- To conduct detailed exploratory data analysis (EDA) to identify patterns and trends.
- To provide a simple, interactive UI for daily operational use.
- To create dynamic Tableau dashboards for discount and profit analysis.
- To understand real life data analysis pipelines used in professional environments.

1.3 Problem Statement

Raw transactional data often contains inconsistencies, missing values, duplicates, and unstructured patterns that make direct analysis difficult. Without proper integration and preprocessing, the datasets cannot reliably support business intelligence. The challenge of this project was to:

- Integrate two separate datasets into one unified analytical table.
- Clean and preprocess the merged data for accuracy.
- Identify meaningful trends related to discount usage, store performance, payment method efficiency, and profitability.
- Present the insights through interactive and interpretable dashboards that can support decision-making.

2.METHODOLOGY

This section describes the complete analytical workflow followed during the internship project. The methodology mirrors professional data analytics pipelines and incorporates structured data integration, processing, and visualization techniques.

2.1 Architecture Overview

The overall architecture of the project follows a streamlined data flow model designed to ensure reliability, accuracy, and analytical depth. The process begins with **data acquisition**, where the two primary datasets—customers and transactional details—were imported into an SQL database. SQL served as the foundation for **data storage**, validation, and relational merging using inner JOIN operations on the InvoiceID field.

The merged output was stored as a SQL VIEW named *MergedData*, improving accessibility for further steps. This integrated dataset was then exported into Python using Jupyter Notebook for cleaning, preprocessing, and exploratory analysis. Python handled inconsistencies in the data, removed duplicates, standardized formats, and generated initial statistical summaries.

After preprocessing, the cleaned dataset was imported into **Tableau**, where interactive dashboards were developed. Tableau facilitated multi-level exploration of discount patterns, profit distribution, store-wise performance, and payment method behavior. This architecture ensured a smooth flow from raw data to analytical insights, maintaining data integrity at each stage.

2.2 Hardware Components

Since this was a software-focused analytical project, the hardware requirements were minimal yet adequate to support SQL operations, Python computation, and Tableau visualization. The project was executed on a personal computer with the following recommended specifications:

- **Processor:** Multi-core CPU (Intel i5/Ryzen 5 or above) to handle SQL queries and Python computations
  RAM: Minimum 8 GB to ensure smooth data processing and responsive Tableau dashboard rendering
- **Storage:** At least 10 GB free disk space for datasets, notebook files, SQL tables, and Tableau workbook files
- **Operating System:** Windows/macOS environment compatible with Tableau and database tools

No external hardware such as sensors or embedded devices was required, as the project focused purely on data analytics and visualization.

2.3 Software Tools

The project utilized multiple software components to complete the data analysis pipeline:

- **SQL (MySQL / PostgreSQL):** Used for loading datasets, data validation, merging tables using JOIN operations, and creating the unified analytical VIEW (*MergedData*).
- **Python (Jupyter Notebook):** Served as the main analytical engine. Libraries such as Pandas and NumPy were used for data cleaning and transformation, while Matplotlib/Seaborn supported initial visual exploration.
- **Tableau:** Used to build interactive dashboards for profit analysis and discount analysis. Tableau's dragand-drop interface enabled drill-down views, filtering, and dynamic insights.
- **CSV Data Files:** The project used *customers.csv* and *details.csv* as inputs, containing transactional and customer-related information.

Together, these tools formed a complete, industry-standard data analytics ecosystem.

3. IMPLEMENTATION

The system implementation includes:

The implementation phase involved transforming the planned methodology into a fully functional data analysis pipeline. Each stage—SQL integration, Python preprocessing, and Tableau visualization—was executed systematically to ensure accuracy, consistency, and clarity in the analytical outcomes.

3.1 SQL Data Integration and Merging

The first step of implementation was importing the two datasets (*customers.csv* and *details.csv*) into the SQL database. Individual SQL tables were created within the *internship* schema, and the raw data was inspected to ensure that fields such as InvoiceID, CustomerID, Price, and TotalSale were correctly imported without corruption or missing entries.

After validating the datasets, the tables were merged using an inner JOIN operation on the shared field *InvoiceID*. This step was crucial for integrating customer payment details with transactional attributes such as product information, store ID, sale amount, and date. The merging query was written as:

To streamline repeated queries, a SQL VIEW named **MergedData** was created. This allowed Python and Tableau to directly retrieve clean, structured, and integrated data from a single source.

3.2 Python-Based Data Cleaning and Analysis

Once the merged data was extracted from SQL, it was loaded into Python using Jupyter Notebook for preprocessing. This step involved several key operations:

- **Duplicate Removal:** Ensured that repeated invoice records did not distort analysis.
- **Missing Value Handling:** Null values in fields like quantity, profit, or discount were addressed using appropriate imputation or removal strategies.
- **Data Standardization:** Converted date formats, corrected numerical data types, and normalized column names for consistency.
- **Derived Features:** New calculated metrics such as total discount contribution or profit ratios were added to enrich analysis.

Exploratory Data Analysis (EDA) was conducted using Pandas, NumPy, Matplotlib, and Seaborn. Various trends, including daily profit variations, store-wise performance, payment method behavior, and discount usage patterns, were identified through descriptive statistics and visual plots.

3.3 Tableau Dashboard Development

The final stage of implementation involved creating interactive dashboards in Tableau. The cleaned dataset was imported into Tableau Desktop, where worksheets were built for key analytical aspects:

- **Profit Analysis Dashboard:** Visualized store-wise profit, payment method profitability, daily revenue trends, and category-level breakdowns.
- **Discount Analysis Dashboard:** Displayed discount distribution across stores, payment methods, and customer segments, along with day-wise discount patterns.

Each dashboard included filters, slicers, bar charts, line charts, and highlight tables to enable dynamic, drill-down exploration. Tableau's interactive environment allowed for intuitive interpretation of complex patterns by business stakeholders.

4. RESULTS AND DISCUSSION

The analysis of the merged customer and transaction datasets produced several meaningful insights regarding store performance, payment behavior, profitability, and discount utilization. By combining SQL integration, Python-based EDA, and Tableau visual dashboards, the project revealed patterns that can support informed business decision-making.

4.1 Payment Method Insights

The results showed clear variations in customer behavior across different payment modes. Wallet payments contributed the highest profit overall, indicating stronger transaction values or higher-margin purchases through this method. Cash payments recorded the highest discount usage, suggesting customers often redeem deals or instore offers when paying in cash. UPI transactions contributed the least profit, reflecting lower purchase amounts or reduced discount activity. These differences highlight the need for payment-specific promotional strategies.

4.2 Store-Wise Performance

Store_C emerged as the top-performing outlet, generating the highest profit among all stores. Its consistency suggests strong customer footfall or efficient sales operations. Store_D showed high discount distribution, which may indicate aggressive promotional activity to attract customers. Stores A and B displayed moderate profits, leaving scope for targeted improvements in inventory, marketing, or customer engagement.

4.3 Daily Trends in Profit and Discount

Day-wise analysis revealed noticeable fluctuations in profit and discount usage across the month. Certain days displayed strong revenue peaks, while others showed dips associated with lower customer activity or reduced promotional campaigns. Discount patterns followed a similar trend, with spikes on specific dates where customer engagement was higher. These observations emphasize the importance of monitoring time-based performance to plan marketing and staffing.

4.4 Overall Interpretation

The combined insights from the dashboards highlight the value of using structured analytics for business evaluation. Store-level differences, payment behavior patterns, and daily fluctuations provide actionable inputs for optimizing pricing strategies, improving store performance, and refining discount policies. Tableau visualizations helped simplify complex data by offering intuitive views that decision-makers can easily interpret.

5. CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This internship project provided a complete, practical experience in implementing a professional data analysis pipeline. By working with SQL, Python, and Tableau, the project demonstrated how raw transactional data can be transformed into meaningful business intelligence. The SQL merging process ensured that both datasets were integrated into a single reliable source, while Python-based cleaning and exploratory analysis helped identify patterns related to profitability, discount usage, payment behavior, and store performance. Tableau added an intuitive layer of visualization, enabling interactive dashboards that clearly communicated insights to nontechnical stakeholders.

Overall, the internship strengthened core analytical skills, improved understanding of data preprocessing techniques, and enhanced the ability to translate complex datasets into actionable insights. The project successfully showcased the importance of structured data workflows and highlighted how organizations can leverage analytics for informed decision-making, performance evaluation, and strategic planning.

5.2 Future Scope

While the current analysis covered essential profit and discount trends, there are several opportunities to expand and improve the system in the future:

- **Integration of predictive models:** Machine learning techniques can be applied to forecast future sales, customer purchasing behavior, or profit trends.
- **Real-time dashboards:** Connecting Tableau directly to a live SQL database would allow continuous updates and real-time monitoring of store performance.
- **Customer segmentation analysis:** Clustering techniques can be used to identify customer groups based on spending habits, discount sensitivity, or payment preferences.
- **Additional metrics:** Incorporating product-level analysis, seasonal trends, or regional comparisons could provide deeper business insights.
- **Automated ETL pipelines:** Tools like Airflow or custom scripts can be used to automate data extraction, transformation, and loading processes for efficiency.

These enhancements would further strengthen the analytical value of the system and support more advanced decision-making processes.

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